Piyush Nargunde

Senior Software Engineer — Java · Kotlin · Python · Agentic AI · Distributed Systems · DevSecOps

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PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Lead with 10+ years of experience delivering engineering-led outcomes across the full technology stack — from service and data tiers through to API and integration layers — in large-scale, regulated, production environments. An AI-native engineer who actively uses agentic development workflows and AI tooling to drive velocity, quality, and force-multiplied outcomes across the team. Strong background in Java, Kotlin, and Python with deep expertise in multi-threaded systems, distributed architectures, microservices, and high-throughput event-driven platforms. Proven track record of delivering commercially focused software end-to-end — from inception and design through deployment and production operation — while uplifting engineering standards and mentoring engineers toward system-level thinking. Comfortable working with ambiguity, taking full ownership of complex work, and influencing technical direction across cross-functional teams.

CORE COMPETENCIES

Java · Kotlin · Multi-threaded Engineering	Agentic AI Workflows & AI-Native Development	Distributed Systems & Microservices
End-to-End Feature & Product Delivery	System Architecture & Solution Design	DevSecOps · CI/CD · Automated Pipelines
Security, Compliance & Audit Engineering	Observability · Monitoring · Alerting · SRE	Event-Driven & Messaging Architectures
AI Tooling Protocols & Governance	Engineering Mentoring & Force Multiplication	Cloud-Native Platforms (AWS · Kubernetes)

WORK EXPERIENCE

Development Lead

2021 – Present

Optus Australia Pty Limited · Sydney, NSW

- Deliver production-grade software end-to-end across Java/Spring Boot, Python, and C++ services — owning the full lifecycle from inception and design through deployment, production operation, and continuous improvement across mission-critical BSS platforms serving millions of subscribers.
- Lead and actively use agentic coding tools and AI-assisted engineering workflows — introducing AI-native development practices that measurably improve team velocity, code quality, and delivery throughput; act as the force multiplier for the team's engineering output.
- Establish and enforce engineering standards, DevSecOps practices, code review processes, and CI/CD automation; conduct regular application security audits, remediate vulnerabilities, and ensure compliance with enterprise security and regulatory requirements.
- Drive architecture and solution design across distributed, event-driven platforms — led cloud migration of 10+ core applications to AWS; designed DR frameworks, AutoHeal capabilities, and technical cutover strategies for business-critical systems.
- Implement observability, monitoring, and alerting frameworks across production platforms; act as technical escalation point for complex incidents and drive root-cause analysis and resolution across infrastructure and application layers.
- Mentor and coach engineers — guiding transition from task-level execution toward architectural ownership and system-level thinking; deliver KPI and OKR reporting to senior stakeholders translating engineering outcomes into business impact.

- Partner closely with Product, Architecture, and Platform teams across cross-functional squads, managing concurrent transformation, BAU, and modernisation initiatives with predictable delivery outcomes.

Advanced Software Engineer

2019 – 2021

IR Limited · Sydney, NSW

- Owned end-to-end delivery of backend features across an enterprise SaaS product — taking commercially shipped features from design and prototyping through implementation, testing, and production deployment in Python (FastAPI), C#, and C++.
- Designed and implemented Predictive Analytics as a commercially released product capability — applying machine learning to drive data-informed customer outcomes; researched and prototyped new product features aligned to market and customer needs.
- Established an engineering framework for SAST code scanning (CodeSonar) fulfilling cybersecurity and compliance obligations — enabling product migration across two technology stacks and remediation of 2,000+ security vulnerabilities.
- Implemented end-to-end observability (logging, metrics, traces) using OpenTelemetry; automated system and unit testing frameworks to enable early bug detection and improve production readiness and release confidence.
- Acted as Scrum Master — driving agile delivery practices, removing blockers, and enabling cross-functional team performance across engineering, product, and design stakeholders.

Software Development Team Lead

2012 – 2019

Amdocs Australia Pty Ltd · Sydney, NSW

- Built and owned production-grade features in Java and C++ for high-volume billing, charging, and CRM platforms — distributed transactional systems processing millions of events daily with strict correctness, availability, and performance requirements.
- Operated at Staff Engineer level contributing to product roadmap and technical direction; liaised with telecom clients across requirements analysis, solution design, and delivery — influencing product priorities and customer outcomes directly.
- Implemented Tweet Sentiment Analysis using NLP (machine learning) enabling real-time customer feedback ingestion and data-driven business decisions; reduced daily E2E bill run from 12 hours to 8 hours through targeted profiling and performance optimisation.
- Developed a billing knowledge base that slashed incident turnaround time by 50%; implemented high-availability improvements in CPQ flows; established reusable engineering frameworks and best practices across the team.

KEY PROJECTS

AI-Native Engineering Transformation

Agentic Coding Tools · AI SDLC · CI/CD Integration · Code Governance · AI Protocols

Optus

Designed and drove adoption of AI-native engineering practices across a multi-product BSS engineering team — introducing agentic coding assistants, AI-integrated SDLC workflows, and governance protocols to ensure AI-generated code met enterprise standards for security, performance, and maintainability. Resulted in measurable improvements to team velocity and code quality; coached engineers to evolve from manual execution toward architectural oversight of AI-assisted delivery.

DevSecOps Pipeline & Security Engineering Framework

GitLab CI · ArgoCD · Docker · Kubernetes · CodeSonar · SAST · Terraform · Shell Scripting

Optus & IR Limited

Designed and implemented CI/CD pipelines and release automation with embedded security scanning (SAST), test coverage gates, and container-based deployment orchestration. At IR Limited, established the code scanning framework that enabled product migration across two stacks and remediated 2,000+ vulnerabilities — fulfilling cybersecurity compliance obligations and significantly improving platform security posture. At Optus, extended this across 10+ BSS applications with automated DR and recovery capabilities.

Observability, Monitoring & SRE Platform

OpenTelemetry · Prometheus · AWS CloudWatch · Logging · Metrics · Distributed Tracing

IR Limited & Optus

Implemented end-to-end observability across production platforms — structured logging, distributed tracing, and metrics instrumentation using OpenTelemetry and Prometheus. Enabled rapid incident diagnosis, performance bottleneck identification, and data-driven continuous improvement across distributed backend services. At Optus, extended with AWS CloudWatch dashboards and alerting to support SRE practices and mission-critical platform operations.

EDUCATION

Bachelor of Engineering, Computer Science

SGSITS, India

Building Scalable Digital Networks

Micro-credential

La Trobe University

Cyber and Data Security

Micro-credential

La Trobe University

CERTIFICATIONS

AWS Certified AI Practitioner
Amazon Web Services

AWS Certified Solutions Architect
Amazon Web Services

Azure Developer Associate
Microsoft

Google Cloud Fundamentals: Core Infrastructure
Google

Academy Accreditation – Generative AI Fundamentals
Databricks

AWS Certified Developer – Associate
Amazon Web Services

Azure Fundamentals
Microsoft

SKILLS

LANGUAGES	Java, Python, Go, C++, C#, Kotlin (working knowledge), Shell Scripting / Bash, SQL, Perl
FRAMEWORKS & APIS	Spring Boot, Spring Batch, FastAPI, REST APIs, Microservices, Helm, Maven
CLOUD & PLATFORM	AWS (EC2, EKS, S3, RDS, VPC, IAM, CloudWatch, ECR, Secrets Manager), Kubernetes, Docker, Kafka, Ansible, ArgoCD, Airflow
DATABASES	PostgreSQL, Oracle, SQLite, MS SQL Server, Data Modelling, SQL Optimisation
OBSERVABILITY & SRE	OpenTelemetry, Prometheus, AWS CloudWatch, Logging / Metrics / Traces, Application Performance Monitoring
DEVSECOPS & CI/CD	GitLab CI, ArgoCD, Terraform, CodeSonar (SAST), Security Audits, Compliance Engineering, Semver
AI & AGENTIC	Agentic Coding Tools, AI-Native SDLC, AI Governance Protocols, Predictive Analytics, NLP, Generative AI (Databricks), AWS AI Practitioner
METHODOLOGY	Agile, Scrum Master, Code Review, System & Solution Design, Technical Leadership, Mentoring, OKR Reporting